
Mixed effects analysis for seizure trigger data

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This script uses a moving average of 90 days in combination with missed doses of ASMs on the last day of the window to look at the relationship with seizures on day window+1 (i.e., day 91 for the first window). This code has also been extended to look at lag terms of ASM adherence over the past 7 days with seizure occurrence. Importantly, as various individuals contribute repeated samples to the dataset (due to the nature of the time windows), it is important to account for this structure, as many typical analyses have the assumption that all the samples are completely independent.

Format data for excel spreadsheet

Load file

```
filename = 'all_patient_diaries.xlsx';

% Extract information about number of sheets in the excel
[~, sheetNames] = xlsfinfo(filename);

% Load the data from each sheet/participant as a separate variable
for i = 1:length(sheetNames)
    sheetData = xlsread(filename, sheetNames{i});
    varName = matlab.lang.makeValidName(sheetNames{i});
    assignin('base', varName, sheetData);
end

% Extract only the seizure data (i.e., column 2) and place it all into one
% variable called seizure where each column is a different participant
for i = 1:length(sheetNames)
    sheetData = xlsread(filename, sheetNames{i});
    seizure(:,i)=sheetData(:,2);
end

% As the raw data is in half days, let's shrink the temporal data such that
% it goes by days rather than half days. The below commands just takes into
% account whether there was a seizure/no seizure for a given day. For
% example, if someone had 3 seizures in a given day, under this format they
% would just be counted as having 1 seizure day for that day.
[numRows, numCols] = size(seizure);
seizure_days = zeros(numRows / 2, numCols);
```

```
for row = 1:numRows/2
    for col = 1:numCols
        % Check if there is a positive number in the two consecutive rows
        if seizure(2*row-1, col) > 0 || seizure(2*row, col) > 0
            seizure_days(row, col) = 1;
        end
    end
end
```

Calculate MA for a 90 day window

Essentially within a 90 day sliding window, this takes the number of seizure days divided by 90 for an estimate of seizure risk.

```
for i = 1:191
    for j = 1:27
        MA(i,j)=sum(seizure_days(i:i+89,j));
        MA_90(i,j)=MA(i,j)/90;
    end
end
```

```
% Since we are looking at the relationship between MA and seizures on day
% sliding window + 1, the first day would be day 91 since the MA requires
% 90 days of data. Let's create a matrix of seizure days that only starts
% at day 91 for further analysis later on.
seizure_days_forecast=seizure_days(91:281,:);
```

Extract the ASM data for each subject

End of data format will be 3D matrix [Dose X ASM X Subject]

```
% Patient 4165: Column 3 Keppra, Column 4 Epidiolex, Column 5 Fluramine
ASM(:,1:3,1)=BIDMC4165(:,3:5);

% Patient 4216: Column 3 CBD oil, Column 4 Lamotrigine
ASM(:,1:2,2)=BIDMC4216(:,3:4);

% Patient 4251: Column 3 CBD oil, Column 5 Onfi
ASM(:,1,3)=BIDMC4251(:,3);
ASM(:,2,3)=BIDMC4251(:,5);

% Patient 4417: Column 5 Keppra, Column 6 Lamictal XR, Column 7 Vimpat
ASM(:,1:3,4)=BIDMC4417(:,5:7);

% Patient 4614: Column 3 Aptiom, Column 4 Xcopri
ASM(:,1:2,5)=BIDMC4614(:,3:4);

% Patient 4684: Column 5 Phenobarb, Column 6 Lamotrigine, Column 8
% Levetiracetam
ASM(:,1:2,6)=BIDMC4684(:,5:6);
ASM(:,3,6)=BIDMC4684(:,8);

% Patient 4711: Column 3 Felbatol, Column 4 Zonisamide, Column 5 Lamictal
```

```
ASM(:,1:3,7)=BIDMC4711(:,3:5);

% Patient 4722: Column 3 Felbatol, Column 4 Onfi, Column 6 Depakote, Column
% 7 Gabapentin, Column 8 Zonisamide, Column 9 Fenfluramine
ASM(:,1:2,8)=BIDMC4722(:,3:4);
ASM(:,3:6,8)=BIDMC4722(:,6:9);

% Patient 4768: Column 3 Topomax, Column 4 Epidiolex, Column 5 Lamictal,
% Column 6 Zarontin
ASM(:,1:4,9)=BIDMC4768(:,3:6);

% Patient 4835: Column 3 Oxtellar, Column 4 Epidiolex, Column 5 Clobazam,
Column 9 Fintepla
ASM(:,1:3,10)=BIDMC4835(:,3:5);
ASM(:,4,10)=BIDMC4835(:,9);

% Patient 4982: Column 3 Lamotrigine, Column 4 Banzel, Column 5 Epidiolex
ASM(:,1:3,11)=BIDMC4982(:,3:5);

% Patient 5057: Column 4 Keppra, Column 6 Onfi
ASM(:,1,12)=BIDMC5057(:,4);
ASM(:,2,12)=BIDMC5057(:,6);

% Patient 5117: Column 4 Zonisamide
ASM(:,1,13)=BIDMC5117(:,4);

% Patient 5146: Column 3 Zonisamide
ASM(:,1,14)=BIDMC5146(:,3);

% Patient 5184: Column 3 Valproic Acid, Column 4 Keppra
ASM(:,1:2,15)=BIDMC5184(:,3:4);

% Patient 5274: Column 3 Lamictal XR, Column 6 Gabapentin, Column 7
% Fycompa
ASM(:,1,16)=BIDMC5274(:,3);
ASM(:,2:3,16)=BIDMC5274(:,6:7);

% Patient 5392: Column 3 Vimpat, Column 4 Keppra, Column 5 Depakote, Column
% 6 Zonegran
ASM(:,1:4,17)=BIDMC5392(:,3:6);

% Patient 5412: Column 3 Zonegran, Column 4 Trileptal, Column 5 Fycompa
ASM(:,1:3,18)=BIDMC5412(:,3:5);

% Patient 5507: Column 3 Briviact, Column 4 Lamotrigine, Column 5
% Epidiolex, Column 6 Xcopri
ASM(:,1:4,19)=BIDMC5507(:,3:6);

% Patient 5606: Column 3 Vimpat, Column 4 Epidiolex, Column 6 Keppra,
% Column 7 Keppra XR, Column 8 XCopri
ASM(:,1:2,20)=BIDMC5606(:,3:4);
ASM(:,3:5,20)=BIDMC5606(:,6:8);

% Patient 5698: Column 3 Keppra, Column 7 Depakote ER, Column 18 XCopri
```

```

ASM(:,1,21)=BIDMC5698(:,3);
ASM(:,2,21)=BIDMC5698(:,7);
ASM(:,3,21)=BIDMC5698(:,18);

% Patient 5708: Column 9 Keppra, Column 14 XCopri, Column 20 Lamotrigine,
% Column 21 Clobazam, Column 26 Epidiolex
ASM(:,1,22)=BIDMC5708(:,9);
ASM(:,2,22)=BIDMC5708(:,14);
ASM(:,3:4,22)=BIDMC5708(:,20:21);
ASM(:,5,22)=BIDMC5708(:,26);

% Patient 5731: Column 3 Carbamazepine, Column 4 Levetiracetam, Column 5
% Vimpat, Column 6 repeat Vimpat?? will ignore for now given no negative
% values/missed doses anyways
ASM(:,1:3,23)=BIDMC5731(:,3:5);

% Patient 5782: Column 3 Vimpat, Column 6 XCopri
ASM(:,1,24)=BIDMC5782(:,3);
ASM(:,2,24)=BIDMC5782(:,6);

% Patient 5790: Column 3 Keppra
ASM(:,1,25)=BIDMC5790(:,3);

% Patient 5799: Column 3 Keppra, Column 5 Tegretol XR
ASM(:,1,26)=BIDMC5799(:,3);
ASM(:,2,26)=BIDMC5799(:,5);

% Patient 59589: Column 3 Topamax, Column 4 Briviact, Column 5 Vimpat
ASM(:,1:3,27)=BIDMC59589(:,3:5);

```

Matrix of missed doses

The raw data for the ASM data is in half days. Let's similarly shrink the temporal resolution such that we look at ASMs in days rather than half days. We will account for missed doses in terms of "administration periods". I.e., if an individual missed morning doses of three different ASMs, that only counts as one missed administration period. For the possible range of values here, for each day it can be 0 (no missed doses), 1 (1 missed administration period, i.e., morning vs night), and 2 (missed both morning and night doses if scheduled for it).

```

[numRows, numCols, numSubs] = size(ASM);
Missed_Doses = zeros(numRows / 2, numCols, numSubs);
    for row = 1:numRows/2
        for col = 1:numCols
            for subs = 1:numSubs
                if ASM(2*row-1, col, subs) < 0 && ASM(2*row, col, subs) < 0
                    Missed_Doses(row, col, subs) = 2;
                elseif ASM(2*row-1, col, subs) < 0 && ASM(2*row, col, subs) >= 0
                || ASM(2*row-1, col, subs) >= 0 && ASM(2*row, col, subs) < 0
                    Missed_Doses(row, col, subs) = 1;
                end
            end
        end
    end
end

```

```
for j = 1:27
    Storage=Missed_Doses(:, :, j);
    Missed_Doses_Periods(:, j)=max(Storage, [], 2);
end

% Since the first 90 days are used to estimate MA, for the missed doses,
% the first number used should be on day 90 to look at how it influences
% seizure risk on day 91. The last day used should be day 280, since day
% 281 is the last day to look at seizure risk.
Missed_Doses_Periods_clean=Missed_Doses_Periods(90:280, :);

% Organize data for mixed-effects analysis with lagged predictors
% Let's organize the data first into a table and concatenate the variables
% vertically. 191 estimates x 27 subjects = 5157.
MAConcat=reshape(MA_90, [5157, 1]); %concatenate the MA data into a vector
SeizureDaysConcat=reshape(seizure_days_forecast, [5157, 1]); %concatenate the
seizure occurrence data
```

Create lag terms for Missed_Doses_Periods_clean

The variable Missed_Doses_Periods_clean represents the missed dose on the day **immediately before** the seizure outcome day. This is Lag 1. We will now create 6 more lags to look back a full week (Lags 2 through 7).

```
num_subjects = 27;
num_estimates = 191;
num_additional_lags = 6; % For days t-2 through t-7

% Rename for clarity and reshape into a single vector
MD_Lag1_Concat = reshape(Missed_Doses_Periods_clean, [5157, 1]);

% Initialize matrix to store the additional lagged data (lags 2 through 7)
MissedDoses_Lagged = NaN(num_estimates, num_subjects, num_additional_lags);

for s = 1:num_subjects
    current_subject_data = Missed_Doses_Periods_clean(:, s);
    for l = 1:num_additional_lags
        % Create the lagged vector for this subject and lag
        % Lag 'l' here corresponds to an actual lag of 'l+1' days.
        % For example, l=1 is Lag 2, l=6 is Lag 7.
        lagged_vector = [NaN(1, 1); current_subject_data(1:end-1)];
        MissedDoses_Lagged(:, s, l) = lagged_vector;
    end
end

% Reshape the lagged matrices into single vectors
MD_Lag2_Concat = reshape(MissedDoses_Lagged(:, :, 1), [5157, 1]);
MD_Lag3_Concat = reshape(MissedDoses_Lagged(:, :, 2), [5157, 1]);
MD_Lag4_Concat = reshape(MissedDoses_Lagged(:, :, 3), [5157, 1]);
MD_Lag5_Concat = reshape(MissedDoses_Lagged(:, :, 4), [5157, 1]);
MD_Lag6_Concat = reshape(MissedDoses_Lagged(:, :, 5), [5157, 1]);
MD_Lag7_Concat = reshape(MissedDoses_Lagged(:, :, 6), [5157, 1]);
```

```
% FEATURE SCALING
% Standardize the MA variable
MA_scaled = (MAConcat - mean(MAConcat)) / std(MAConcat);
% Standardize the missed dose and its lagged variables
MD_Lag1_scaled = (MD_Lag1_Concat - mean(MD_Lag1_Concat, 'omitnan')) /
std(MD_Lag1_Concat, 'omitnan');
MD_Lag2_scaled = (MD_Lag2_Concat - mean(MD_Lag2_Concat, 'omitnan')) /
std(MD_Lag2_Concat, 'omitnan');
MD_Lag3_scaled = (MD_Lag3_Concat - mean(MD_Lag3_Concat, 'omitnan')) /
std(MD_Lag3_Concat, 'omitnan');
MD_Lag4_scaled = (MD_Lag4_Concat - mean(MD_Lag4_Concat, 'omitnan')) /
std(MD_Lag4_Concat, 'omitnan');
MD_Lag5_scaled = (MD_Lag5_Concat - mean(MD_Lag5_Concat, 'omitnan')) /
std(MD_Lag5_Concat, 'omitnan');
MD_Lag6_scaled = (MD_Lag6_Concat - mean(MD_Lag6_Concat, 'omitnan')) /
std(MD_Lag6_Concat, 'omitnan');
MD_Lag7_scaled = (MD_Lag7_Concat - mean(MD_Lag7_Concat, 'omitnan')) /
std(MD_Lag7_Concat, 'omitnan');

%Let's create a subject ID that corresponds to each of the entries above
for i = 1:num_subjects;
    SubjectID(1:num_estimates,i)=i;
end
SubjectIDConcat=reshape(SubjectID,[5157,1]);

% Place the data into a single table, using the scaled variables
tbl = table(MD_Lag1_scaled,MD_Lag2_scaled,MD_Lag3_scaled,MD_Lag4_scaled, ...
    MD_Lag5_scaled,MD_Lag6_scaled,MD_Lag7_scaled, ...
    MA_scaled,SubjectIDConcat,SeizureDaysConcat,'VariableNames',
    {'MD_Lag1','MD_Lag2', ...
    'MD_Lag3','MD_Lag4','MD_Lag5','MD_Lag6','MD_Lag7','MA_tbl','ID_tbl', ...
    'Seizure_tbl'});

% Find the rows that have NaN values in any of the lag columns
% This ensures that all models are fitted on the same number of observations
non_nan_rows = ~any(ismissing(tbl(:, {'MD_Lag1', 'MD_Lag2', 'MD_Lag3', ...
    'MD_Lag4', 'MD_Lag5', 'MD_Lag6', 'MD_Lag7'}))), 2);
tbl_clean = tbl(non_nan_rows, :);
fprintf('Number of observations for all models: %d\n\n', size(tbl_clean, 1));
```

Fit models for different combinations of lag terms and store them

Models will be built cumulatively: Model 0: Intercept-only (Null) Model 1: MA only Model 2: MA + MD_Lag1 (missed dose day before) Model 3: MA + MD_Lag1 + MD_Lag2 ... and so on up to Lag 7.

```
% Initialize a cell array to store the fitted model objects
all_models_lme = cell(num_additional_lags + 3, 1); % Null, MA-only, MA+Lag1,
plus 6 more
% New array to store the p-values of the overall deviance tests
overall_p_values = zeros(num_additional_lags + 2, 1);
```

```
% Fit the intercept-only model (the true null model)
formula_null = 'Seizure_tbl ~ 1 + (1 | ID_tbl)';
lme_null = fitglme(tbl_clean, formula_null, 'Distribution', 'Binomial');
all_models_lme{1} = lme_null;
deviance_null = -2 * lme_null.LogLikelihood;
fprintf('Model 0 (Intercept Only): Deviance = %.4f\n\n', deviance_null);

% Fit Model 1: MA and random effects
formula_ma_only = 'Seizure_tbl ~ MA_tbl + (1 | ID_tbl)';
lme_ma_only = fitglme(tbl_clean, formula_ma_only, 'Distribution',
'Binomial');
all_models_lme{2} = lme_ma_only;
deviance_ma = -2 * lme_ma_only.LogLikelihood;
chi2_stat_ma = deviance_null - deviance_ma;
p_value_ma = 1 - chi2cdf(chi2_stat_ma, 1); % 1 df
overall_p_values(1) = p_value_ma;
fprintf('Model 1 (MA Only):\n');
fprintf(' Formula: %s\n', formula_ma_only);
fprintf(' --- Overall Model Significance (vs. Null) ---\n');
fprintf(' Chi-squared = %.4f, p = %.4f\n', chi2_stat_ma, p_value_ma);
fprintf(' Deviance = %.4f\n\n', deviance_ma);

% Fit Model 2: MA + MD_Lag1 (missed dose on day before)
formula_ma_lag1 = 'Seizure_tbl ~ MA_tbl + MD_Lag1 + (1 | ID_tbl)';
lme_ma_lag1 = fitglme(tbl_clean, formula_ma_lag1, 'Distribution',
'Binomial');
all_models_lme{3} = lme_ma_lag1;
deviance_ma_lag1 = -2 * lme_ma_lag1.LogLikelihood;
chi2_stat_ma_lag1 = deviance_null - deviance_ma_lag1;
p_value_ma_lag1 = 1 - chi2cdf(chi2_stat_ma_lag1, 2); % 2 df (MA + MD_Lag1)
overall_p_values(2) = p_value_ma_lag1;
fprintf('Model 2 (MA + MD_Lag1):\n');
fprintf(' Formula: %s\n', formula_ma_lag1);
fprintf(' --- Overall Model Significance (vs. Null) ---\n');
fprintf(' Chi-squared = %.4f, p = %.4f\n', chi2_stat_ma_lag1,
p_value_ma_lag1);
fprintf(' Deviance = %.4f\n\n', deviance_ma_lag1);

% Loop through and add the remaining lag terms one at a time (Lags 2 through
7)
for i = 1:num_additional_lags
    % Dynamically build the formula string, always including MA and MD_Lag1
    lagged_terms = '';
    for j = 1:i
        % Start adding terms from MD_Lag2 onwards
        lagged_terms = strcat(lagged_terms, sprintf(' + MD_Lag%d', j + 1));
    end
    current_formula = strcat('Seizure_tbl ~ MA_tbl + MD_Lag1', lagged_terms,
' + (1 | ID_tbl)');
    fprintf('Fitting Model %d: %s\n', i + 2, current_formula);

    % Fit the model on the cleaned table
    lme = fitglme(tbl_clean, current_formula, 'Distribution', 'Binomial');
```

```
% Store the fitted model object
all_models_lme{i + 3} = lme;

% Perform the Likelihood Ratio Test (LRT) by comparing to the null model
deviance_full = -2 * lme.LogLikelihood;
chi2_stat = deviance_null - deviance_full;
df = i + 2; % i additional MD parameters + MA + MD_Lag1
p_value = 1 - chi2cdf(chi2_stat, df);

% Store the calculated p-value
overall_p_values(i + 2) = p_value;

fprintf(' --- Overall Model Significance (vs. Null) ---\n');
fprintf(' Chi-squared = %.4f, p = %.4f\n', chi2_stat, p_value);
fprintf(' Deviance = %.4f\n\n', deviance_full);
end

% Display a summary of all fitted models
fprintf('\n--- Summary of All Fitted Models ---\n');
for i = 1:length(all_models_lme)
    fprintf('\nModel %d:\n', i - 1);
    disp(all_models_lme{i});
end

% Display the stored overall p-values
fprintf('\n--- Overall P-values for Each Model (Compared to Intercept-Only Null Model) ---\n');
fprintf('Model 1 (MA only): p = %.4f\n', overall_p_values(1));
fprintf('Model 2 (MA + MD_Lag1): p = %.4f\n', overall_p_values(2));
for i = 1:num_additional_lags
    fprintf('Model %d (MA + Lags 1 to %d): p = %.4f\n', i + 2, i + 1,
overall_p_values(i + 2));
end
```

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